Type-I calibration of fixed-effect interaction tests in linear mixed models: a staged diagnosis of working-covariance mis-specification and a cluster-robust remedy

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Contents

Abstract

Background. The linear mixed model is the standard tool for testing a fixed-effect interaction in longitudinal and clustered data. The model-based Wald test of that interaction is exact only when the working within-cluster covariance is correctly specified. In practice an analyst chooses a convenient parametric residual structure, a continuous-time autoregressive correlation, compound symmetry, or an unstructured form, and then trusts the model-based standard error. The direction and the magnitude of the mis-calibration that follows when the working covariance is wrong are rarely checked.

Methods. We set out a staged diagnostic protocol, a four-channel decomposition of the test statistic, a working-covariance model ladder, and a sample-size gradient, and apply it to a worked example: the test of a biomarker-by-treatment interaction in an aggregated N-of-1 carryover simulation, analysed with an `lme` model carrying a random intercept and a corCAR1 residual correlation.

Results. At the null the model-based standard error of the interaction is over-estimated by 6 to 10 percent, and the realised type-I error is approximately 0.03 against a nominal 0.05. The degrees-of-freedom rule, the positive-definiteness correction, and point-estimate bias are excluded as causes. The model ladder localises the mis-calibration to the corCAR1 residual layer: a random intercept alone is well calibrated. The sample-size gradient shows the bias is asymptotic, not a moderate-sample artefact: it does not attenuate from $N = 35$ to $N = 280$. A CR2 bias-reduced cluster-robust standard error restores the standard-error ratio to approximately one and the type-I error to approximately 0.05. At an alternative cell with a non-null effect the recalibration raises power by a near-uniform increment and leaves the ranking of analysis specifications unchanged.

Conclusions. A mis-specified parametric working covariance can bias the model-based standard error of a mixed-model interaction test asymptotically, in either direction. Cluster-robust standard errors are a general remedy, and the staged diagnostic is a reusable instrument for detecting the problem.

1 Introduction

The linear mixed model is, by a wide margin, the most common vehicle for testing whether an effect is moderated by a covariate in longitudinal or clustered data[1,2]. The quantity of interest is a fixed-effect interaction coefficient, and the inference almost always proceeds through a Wald test: the coefficient estimate is divided by a model-based standard error and referred to a t distribution. That procedure is exact only under a condition that is easy to state and easy to forget. The working covariance, the random-effects structure together with the parametric residual correlation the analyst has assumed, must be correct.

It has been understood since the work of Eicker, Huber, and White that this condition is more demanding than it appears[3–5]. When the working covariance is mis-specified, the model-based standard error is no longer a consistent estimator of the true sampling variability of the fixed-effect estimate. Liang and Zeger[6] carried this insight into longitudinal data analysis: their generalized estimating equations decouple the point estimate, computed under a convenient working correlation, from the variance estimate, computed by an empirical sandwich that remains consistent whatever the working correlation. The sandwich, or cluster-robust, variance estimator has since become a standard instrument, and a substantial literature has refined it for the small-sample case[7–10].

Yet in routine mixed-model practice the working covariance is still trusted. An analyst fitting longitudinal data with `nlme` or a comparable package will choose a residual correlation, very often a continuous-time first-order autoregressive form, because it is pharmacokinetically or biologically plausible[11], and will then report the model-based standard error of the interaction without asking what happens if the chosen form is not exactly the form the data follow. This is not negligence so much as a reasonable working assumption that is seldom audited. But how large is the error when the assumption fails? Is the resulting test conservative or anti-conservative? Does the mis-calibration diminish as the sample grows, so that it can be dismissed as a small-sample nuisance, or does it persist? These questions do not have a single answer, because the answer depends on the interaction being tested and on the way the working covariance departs from the truth. They have to be asked of a particular analysis.

This paper does exactly that for one analysis, and in doing so offers two contributions that we believe are more general than the single case. The first is a diagnostic protocol. We decompose the Wald statistic into four channels through which mis-calibration can enter, and we add two further instruments, a working-covariance model ladder that localises the cause and a sample-size gradient that separates a finite-sample artefact from an asymptotic one. The protocol is not specific to the example and can be applied to any mixed-model interaction test whose calibration is in doubt. The second contribution is the finding the protocol produced. In our worked example the model-based interaction test is conservative, not anti-conservative as the sandwich literature's emphasis on under-estimated standard errors might lead one to expect; the conservatism is material, on the order of ten percent in standard-error terms; and it is asymptotic, persisting undiminished across an eightfold range of sample size. We trace it to a single component of the working covariance, and we show that a cluster-robust standard error removes it.

The worked example is drawn from the pmsimstats-ng project, a simulation study of power to detect biomarker-treatment interactions in aggregated N-of-1 trials under carryover[12]. The example is incidental to the argument. The phenomenon, a parametric working covariance that biases a fixed-effect interaction test asymptotically, and the remedy are not properties of that simulation; they are properties of the mixed-model Wald test, and the example serves only to make them concrete and quantified. We begin in Section 2 with the diagnostic protocol. Section

3 describes the worked example. Section 4 reports the staged diagnosis and the cluster-robust remedy. Section 5 discusses the generality of the result and its limits.

2 A staged diagnostic protocol

Consider a fixed-effect interaction coefficient θ estimated by $\hat{\theta}$, with a model-based standard error $\widehat{\mathrm{SE}}$ and an assigned degrees of freedom ν. The Wald test rejects the null $H_0 : \theta = 0$ at level α when the statistic $T = \hat{\theta}/\widehat{\mathrm{SE}}$ exceeds the critical value $t_{\nu,1-\alpha/2}$ in absolute value. The test is correctly calibrated when the realised rejection rate under H_0 equals α. When it does not, the discrepancy must enter through one of exactly four channels, and the channels are separable by simulation.

2.1 The four-channel decomposition

We begin by writing the interaction estimate in a form that exposes the four channels. Let

$$\hat{\theta} \;=\; \mu + \sigma\, Z^\star, \qquad \mu = \mathrm{E}(\hat{\theta}), \quad \sigma = \mathrm{SD}(\hat{\theta}),$$

so that $Z^\star = (\hat{\theta} - \mu)/\sigma$ is the standardised estimate, of mean zero and unit variance by construction; when $\hat{\theta}$ is approximately normal, $Z^\star$ is approximately standard normal. Write $\overline{\widehat{\mathrm{SE}}}$ for the mean of the model-based standard error and $r = \overline{\widehat{\mathrm{SE}}}/\sigma$ for its ratio to the true sampling standard deviation σ. Then the Wald statistic, to the order at which the variability of $\widehat{\mathrm{SE}}$ about its mean may be neglected, is

$$T \;=\; \frac{\hat{\theta}}{\overline{\widehat{\mathrm{SE}}}} \;\approx\; \frac{\mu + \sigma Z^\star}{r\,\sigma} \;=\; \frac{1}{r}\,(Z^\star + \delta), \qquad \delta = \frac{\mu}{\sigma},$$

where δ is the bias expressed in standard-deviation units. The realised type-I error follows at once,

$$\alpha_{\text{real}} \;=\; \Pr\!\big(|T| > t_{\nu,\,1-\alpha/2}\big) \;\approx\; \Pr\!\big(|Z^\star + \delta| > r\, t_{\nu,\,1-\alpha/2}\big),$$

and exactly four quantities appear in it. They are the four channels: the standardised bias δ, which decentres the statistic; the scale r, which multiplies the rejection threshold; the degrees of freedom ν, which fix the critical value $t_{\nu,1-\alpha/2}$; and the distribution of $Z^\star$, the shape of the reference. When all four take their nominal values, $\delta = 0$, $r = 1$, $\nu = \infty$, and $Z^\star \sim \mathrm{N}(0, 1)$, the expression collapses to $\Pr(|\mathrm{N}(0, 1)| > z_{1-\alpha/2}) = \alpha$ and the test is exact; a departure in any one channel mis-calibrates it. A second identity is worth recording, since the diagnosis turns on it: because $T \approx (Z^\star + \delta)/r$, the dispersion of the statistic satisfies

$$\mathrm{SD}(T) \;\approx\; \mathrm{SD}(Z^\star)\,/\,r,$$

so a statistic standard deviation below one is read at once as a scale inflation ($r > 1$), a sub-normal reference shape, or both.

We measure the four channels by simulation as follows.

1. **Point-estimate bias** (δ). If $\mathrm{E}(\hat{\theta}) \neq 0$ under H_0 the statistic is decentred. We measure μ, and hence $\delta = \mu/\sigma$, by the mean of $\hat{\theta}$ across null replicates.
2. **Standard-error calibration** (r). If the model-based $\widehat{\mathrm{SE}}$ does not estimate the true sampling standard deviation of $\hat{\theta}$, the statistic is rescaled. We measure r as the mean model-reported standard error over the empirical standard deviation of the estimate across replicates. A ratio above one indicates an over-stated standard error and a conservative test; a ratio below one the reverse.
3. **Degrees of freedom** (ν). If ν is too small the t reference has tails too heavy and the test is conservative; if too large, the reverse. We measure this by recomputing the p-values against a normal reference and comparing the realised type-I error.
4. **Reference-distribution shape** (the law of $Z^\star$). Even with $\delta = 0$, $r = 1$, and ν correct, $Z^\star$ may not be standard normal. We measure this by the standard deviation of T under H_0, which should be approximately one, and by the empirical distribution function of the p-values, which should be $\mathrm{Uniform}(0, 1)$.

The individual channels are not new. The bias of the estimate and the ratio of the model-based to the empirical standard error are among the standard performance measures of a simulation study, as codified by Morris, White, and Crowther[13], who recommend that the model-based and empirical standard errors be reported and compared as a matter of course; the degrees of freedom of the reference t are the subject of the small-sample literature of Satterthwaite[14] and Kenward and Roger[15]; and the uniformity of the null p-value distribution is the elementary calibration check that follows from the probability integral transform. What the decomposition contributes is not a new measure but the observation that these four quantities, and exactly these four, compose into the single expression for α_{real} derived above. Two consequences follow. First, the channels are exhaustive: a Wald test is a centred statistic, a scale, a reference shape, and a degrees-of-freedom choice, and the realised error rate is a function of those and of nothing else, so a discrepancy absent from all four cannot exist. Second, because the channels enter the expression in distinct and non-interacting roles, measuring all four does not merely detect a mis-calibration but localises it, identifying which term carries the fault. Since the channels have different causes and different remedies, a diagnosis is in this sense a decomposition and not a checklist.

2.2 The working-covariance model ladder

The four-channel decomposition says *how* a test is mis-calibrated. It does not say *why*. When the decomposition points to the standard-error channel, it is worth being precise about what the scale r measures. Let the interaction estimate be written $\hat{\theta} = \ell' \hat{\beta}$, with ℓ the contrast vector that selects the interaction coefficient from the fixed-effects vector. A fixed-effect estimator that solves a generalized least squares system under a working covariance V_0, when the data are in truth generated under a covariance V, has a model-based variance

$$v_0 \;=\; \ell'\big(X'V_0^{-1}X\big)^{-1}\ell,$$

while its true sampling variance is the sandwich form

$$v \;=\; \ell'\big(X'V_0^{-1}X\big)^{-1}\big(X'V_0^{-1}VV_0^{-1}X\big)\big(X'V_0^{-1}X\big)^{-1}\ell,$$

where X is the fixed-effects design and, the clusters being independent, every cross-product is a sum over clusters, $X'V_0^{-1}X = \sum_i X_i'V_{0i}^{-1}X_i$ and likewise $X'V_0^{-1}VV_0^{-1}X = \sum_i X_i'V_{0i}^{-1}V_iV_{0i}^{-1}X_i$.[5,6] The scale of Section 2.1 is the ratio of these two quadratic forms, $r^2 = v_0/v$. They coincide when $V_0 = V$, and for a generic contrast ℓ they do not otherwise; the scale channel is therefore not a finite-sample nuisance but a direct consequence of the working covariance: r departs from one whenever $V_0 \neq V$, and no quantity of data removes the discrepancy.

This is also what makes a model ladder informative. Each rung is a different working covariance V_0 fitted to the same data, so tracking r along a graded sequence of working covariances, from the simplest to the production model, isolates the rung, that is the covariance component, at which r departs from one. Because every rung is fitted to the identical simulated data, the comparison is free of Monte Carlo noise between models.

2.3 The sample-size gradient

A standard-error mis-calibration may be one of two qualitatively different things. It may be a finite-sample artefact, in which case it attenuates as the sample grows and the working-covariance parameters are estimated more sharply; the small-sample corrections of Kenward and Roger[15,16] address mis-calibration of this kind. Or it may be asymptotic, a consequence of a working covariance that does not match the data-generating covariance no matter how much data is collected, in which case it persists. The two are distinguished by running the decomposition across a range of sample sizes. Attenuation toward nominal calibration marks the finite-sample case; a flat profile marks the asymptotic case. The distinction matters, because a finite-sample artefact at a realistic sample size may be tolerable, whereas an asymptotic bias is a property of the method and will not be outgrown.

3 Worked example: an interaction test under carryover

The example is the test of a biomarker-by-treatment interaction in the carryover-sensitivity simulation study of the pmsimstats-ng project. Aggregated N-of-1 trials repeatedly cross each participant between an active treatment and a control condition, and the scientific question is whether a baseline biomarker moderates the treatment effect. Carryover, the persistence of drug effect into a washout period, complicates the analysis, and the study compares analysis-model strategies for accommodating it[13]. The reporting follows the ADEMP structure of Morris, White, and Crowther[13].

3.1 The analysis model

For a symptom score $S_{x,it}$ on participant i at timepoint t, the analysis model is the linear mixed model

$$S_{x,it} \;=\; \beta_0 + \beta_1\,\mathrm{bm}_i + \beta_2\,t + \beta_3\,X_{it} + \theta\,(\mathrm{bm}_i \times X_{it}) + b_i + \varepsilon_{it},$$

fitted by `nlme::lme` with a participant random intercept $b_i \sim \mathrm{N}(0, \sigma_b^2)$ and a continuous-time first-order autoregressive residual correlation, corCAR1(form $= \sim t \mid$ ptID). Here bm_i is the standardised biomarker, X_{it} a drug-exposure predictor, and θ the interaction coefficient of interest. The continuous-time autoregressive residual was adopted in the project for its clinical realism, nearby measurements being more strongly correlated than distant ones, and for the numerical range it affords[11]. The Wald test of θ is the object of this paper.

3.2 The null configuration

Calibration is a null-hypothesis property, so the diagnosis is run where the biomarker exerts no moderation, $c_{bm} = 0$, and the true interaction is therefore $\theta = 0$. The primary cell is the reference configuration of the parent study: the Hybrid N-of-1 design, $N = 70$ participants, an exponential carryover form with a one-week half-life, and an autoregressive parameter of 0.7. A preliminary check established that at $c_{bm} = 0$ the study's two data-generating architectures collapse to a bit-identical process, so the diagnosis need not stratify by architecture.

The feature of the example that matters for the argument is the relationship between the working covariance and the truth. The simulated symptom score is a sum of three latent factors, each following its own autoregressive process, with timepoint-dependent variances, plus a participant-level baseline term. The working covariance of the analysis model, one random intercept and one single-parameter corCAR1 residual, cannot reproduce that structure exactly. The example is therefore an instance of the general situation the protocol is designed for: a parametric working covariance that is plausible, conventional, and not exactly correct.

4 Results

The diagnosis was carried out at 5000 null replicates for the four-channel decomposition and the model ladder, and at 1500 replicates per point for the sample-size gradient. The cluster-robust remedy was verified at 3000 replicates. A rejection rate estimated from n_{sim} replicates carries a Monte Carlo standard error of $\sqrt{p(1-p)/n_{\mathrm{sim}}}$[13]; at a type-I error near 0.03 this is approximately 0.0024 at 5000 replicates, 0.0031 at 3000, and 0.0044 at 1500, and the comparisons to the nominal level reported below are read against those tolerances. All replicate-level outputs

Table 1: Four-channel decomposition at the primary null cell (Hybrid design, $N = 70$, $c_{bm} = 0$, 5000 replicates), for the three analysis specifications A1, A2, A3. The interaction estimate is effectively unbiased; the SE ratio exceeds one; the statistic is correspondingly compressed; the t and z references coincide because the degrees of freedom are large.

Spec	Mean estimate	SE ratio r	SD(T)	Type I (t)	Type I (z)	Mean DF
A1	+0.032	1.069	0.932	0.033	0.034	487
A2	+0.039	1.097	0.909	0.029	0.029	487
A3	+0.033	1.061	0.939	0.035	0.036	486

are retained for inspection.

4.1 The conservatism is standard-error over-estimation

Table 1 reports the decomposition for the three analysis specifications carried by the parent study, which differ only in the drug-exposure predictor X_{it} and are treated here as three instances of the same interaction test. The reading is consistent across all three. The interaction estimate is effectively unbiased: the null mean falls between +0.03 and +0.04 across the three specifications, on a coefficient whose sampling standard deviation is near unity, a displacement of about 0.035 of a standard deviation. A displacement of that size is statistically detectable at 5000 replicates but is negligible in magnitude, and in any case it points toward rejection and so cannot produce a conservative test. The degrees-of-freedom channel is excluded directly: the model assigns approximately 487 degrees of freedom to the interaction term, and recomputing the p-values against a normal reference moves the realised type-I error by about 0.001, as the near-identical t- and z-reference columns show.

The discrepancy is in the standard-error channel. The model-based standard error exceeds the empirical sampling standard deviation of the estimate by 6 to 10 percent, and the standard deviation of the test statistic agrees with the reciprocal of that ratio to within half a percent, as the identity $\mathrm{SD}(T) \approx \mathrm{SD}(Z^\star)/r$ with $\mathrm{SD}(Z^\star) = 1$ leads us to expect. With the other three channels inert, $\delta \approx 0$, ν large, and $Z^\star$ standard normal, the realised type-I error of Section 2.1 reduces to the closed form

$$\alpha_{\text{real}} \;\approx\; 2\,\Phi\!\big(-r\,t_{\nu,\,1-\alpha/2}\big),$$

the probability that a standard normal exceeds the threshold $r\,t_{\nu,1-\alpha/2}$ rather than $z_{1-\alpha/2}$. Differentiating this closed form exposes the leverage of the scale channel. Near $r = 1$, with ν large enough that $t_{\nu,1-\alpha/2} \approx z_{1-\alpha/2} = 1.96$,

$$\frac{\mathrm{d}\,\alpha_{\text{real}}}{\mathrm{d}\,r} \;=\; -2\,t_{\nu,1-\alpha/2}\,\phi\!\big(r\,t_{\nu,1-\alpha/2}\big) \;\approx\; -0.23,$$

so to first order $\alpha_{\text{real}} \approx 0.05 - 0.23\,(r-1)$: the realised error falls roughly 2.3 times as fast as the standard error is inflated. A six-to-ten percent over-statement of the standard error therefore depresses the type-I error into the low 0.03 range, which is the sense in which a numerically modest mis-calibration of the standard error is a material mis-calibration of the test. For the standard-error ratios of Table 1 this predicts a type-I error of 0.031 to 0.037 across the three specifications, matching the observed 0.029 to 0.035 (at $\alpha = 0.05$) to within Monte Carlo error. The scale channel alone, with no contribution from the other three, reproduces the conservatism.

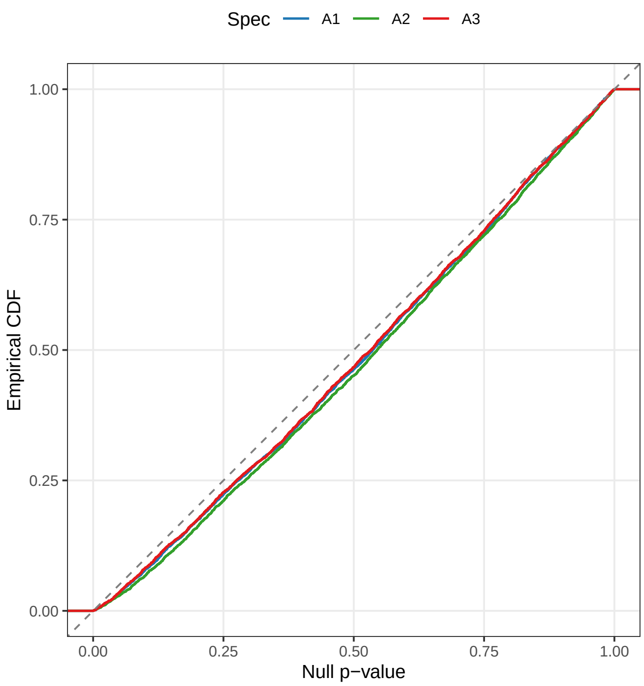

Figure 1: Empirical distribution function of the null p-values at the primary cell, against the Uniform(0,1) diagonal. All three specifications bow uniformly below the diagonal, the signature of a global scale compression rather than a tail anomaly.

Figure 1 confirms the nature of the mis-calibration. The empirical distribution function of the null p-values lies below the diagonal across the whole unit interval and for all three specifications. A test that failed only in its tails would depart from the diagonal only near zero; the uniform bowing seen here is the signature of a global scale compression, which is what an inflated standard error produces. The positive-definiteness correction available to the data generator was not triggered in any of the cell's paths and is excluded as a contributor.

4.2 The corCAR1 residual layer is the cause

The decomposition says the standard error is over-estimated. The model ladder, Table 2, says which part of the model does the over-estimating. Four working covariances were fitted to the identical null datasets: ordinary least squares with no within-cluster term (M0), a random intercept alone (M1), a corCAR1 residual alone (M2), and the production combination of a random intercept with a corCAR1 residual (M3).

Table 2: Working-covariance model ladder at the primary null cell (5000 replicates), for specifications A1 and A2. The same datasets are refitted under each structure. M1, a random intercept alone, is well calibrated; M3, the production model, adds a corCAR1 residual and is conservative.

Model	Structure	r (A1)	r (A2)	Type I (A1)	Type I (A2)
M0	OLS, no within-cluster correlation	1.323	1.289	0.010	0.012
M1	Random intercept only	0.989	0.972	0.049	0.055
M2	corCAR1 residual only	1.023	1.070	0.043	0.036
M3	Random intercept + corCAR1 (production)	1.068	1.097	0.033	0.029

The result is unambiguous. M1, the random intercept with no residual correlation, is well calibrated: its standard-error ratio sits between 0.97 and 0.99 and its type-I error between 0.049 and 0.055. M3, which adds the corCAR1 residual on top of that random intercept, is conservative: the ratio rises to between 1.07 and 1.10 and the type-I error falls to between 0.029 and 0.033. The corCAR1 residual-correlation layer is therefore the proximate cause of the conservatism. It is worth being precise about what this does and does not say. It does not say that the corCAR1 residual is wrong in any absolute sense; the simulated factors are genuinely autocorrelated, so a residual correlation term is capturing a real feature of the data. It says that adding that term to a model that already carries a random intercept degrades the calibration of the fixed-effect interaction test.

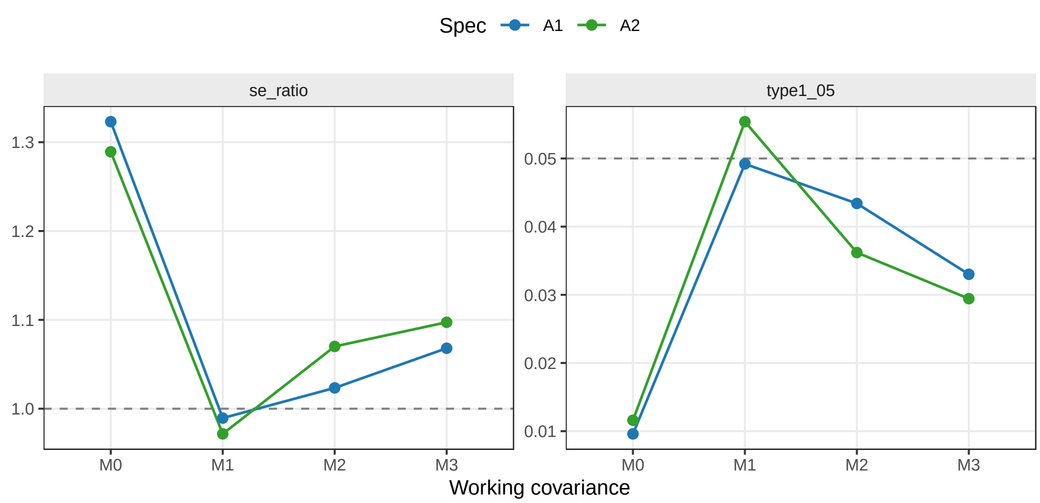

Figure 2: Interaction standard-error calibration across the model ladder. Left, the ratio of model-reported to empirical standard error; right, the realised type-I error. The dashed lines mark perfect calibration. The random intercept alone (M1) sits on both reference lines; the production model (M3) does not.

Two further features of the ladder, visible in Figure 2, are worth recording. The first is that M0, ordinary least squares, is not anti-conservative, as the familiar warning that ignoring positive within-cluster correlation under-states standard errors would suggest. It is strongly conservative, with a standard-error ratio near 1.3. The reason is instructive: the interaction here is a within-participant

Table 3: Sample-size gradient under the production model M3 (1500 replicates per point). The standard-error ratio r and the type-I error (T1) are shown for each specification. Neither trends toward its nominal value as N increases.

N	r A1	r A2	r A3	T1 A1	T1 A2	T1 A3
35	1.072	1.103	1.064	0.038	0.028	0.039
70	1.056	1.083	1.050	0.043	0.031	0.045
140	1.086	1.098	1.079	0.041	0.033	0.042
280	1.052	1.070	1.046	0.037	0.028	0.035

contrast, and ordinary least squares charges it the full residual variance, between-participant and within-participant combined, when the error relevant to a within-participant contrast is only the within-participant part. In the idealised case of a pure within-participant contrast with exchangeable errors the inflation is the variance ratio $r^2_{\mathrm{OLS}} \approx (\sigma_b^2 + \sigma_\varepsilon^2)/\sigma_\varepsilon^2$, the total residual variance over its within-participant component, exceeding one by the random-intercept variance σ_b^2. The familiar warning, the Moulton pitfall, concerns between-cluster effects[17]; for a within-cluster effect the sign of the OLS error is reversed. The second feature is that the corCAR1 residual without a random intercept (M2) is intermediate. Taken together the ladder shows that calibration is recovered by the random intercept alone and removed by the corCAR1 addition, and that the magnitude and even the sign of the mis-calibration is a property of the working covariance, not a universal constant.

4.3 The conservatism is asymptotic, not a small-sample artefact

If the conservatism were a finite-sample artefact, it would diminish as the sample grew and the working-covariance parameters were estimated more precisely. Table 3 and Figure 3 show that it does not. Across an eightfold range of sample size, from $N = 35$ to $N = 280$ participants, the standard-error ratio holds between 1.05 and 1.10 with no trend toward one, and the type-I error holds between 0.028 and 0.045 with no trend toward 0.05. At the largest sample size the production specification still rejects the true null at 0.028. The conservatism is asymptotic. It is a property of the M3 working covariance and will not be outgrown.

The explanation that is consistent with all three results is a working-covariance mismatch rather than a small-sample effect. The simulated symptom score is a mixture of autoregressive factors with unequal, timepoint-dependent variances, plus a participant baseline term. The M3 working covariance, a single random intercept and a single-parameter corCAR1 residual, cannot match a mixture of autoregressive processes exactly. Increasing the number of participants estimates the parameters of that working covariance ever more sharply, but it does not change the fact that the covariance being estimated is the wrong one, and so the model-based standard error converges to a biased target. The bias persists because it is a bias of the model, not of the estimate of the model.

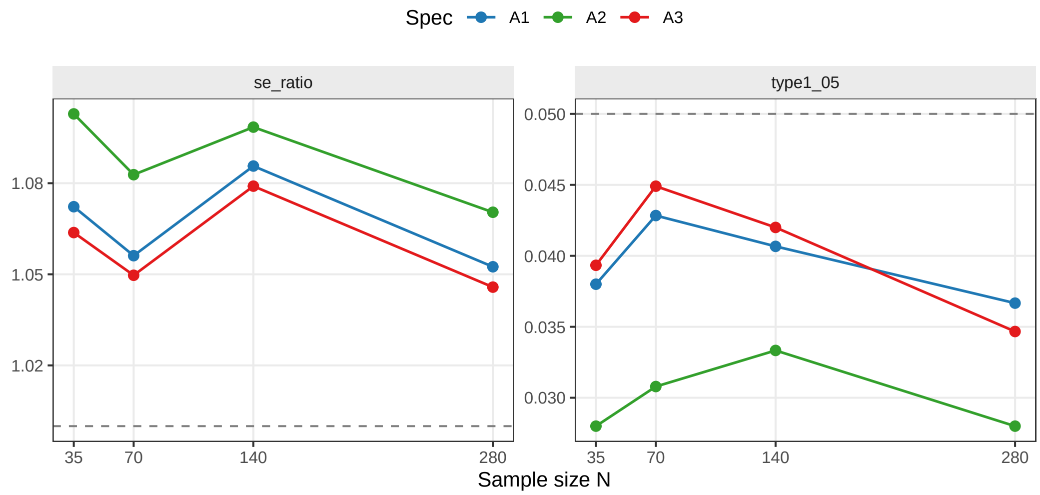

Figure 3: Type-I error and standard-error ratio against sample size under the production model M3. Neither panel trends toward its dashed reference line as the number of participants increases from 35 to 280.

Table 4: Cluster-robust remedy at the primary null cell (3000 replicates). The model-based standard error of the production model is replaced by a CR2 bias-reduced cluster-robust standard error clustered on participant, with Satterthwaite degrees of freedom. The CR2 standard-error ratio is approximately one and the type-I error approximately 0.05.

Spec	r model-based	r CR2	Type I model	Type I CR2	CR2 df
A1	1.062	0.978	0.036	0.048	23.7
A2	1.096	1.000	0.029	0.045	23.7
A3	1.053	0.976	0.039	0.049	23.7

4.4 A cluster-robust standard error restores calibration

The diagnosis points to a remedy that does not require abandoning the corCAR1 model. If the model-based standard error is biased because the working covariance is wrong, the natural replacement is a standard error that is consistent whatever the working covariance[5,6,18]. We replace the model-based standard error of the interaction with a cluster-robust sandwich standard error, clustered on participant, in the bias-reduced CR2 form of Bell and McCaffrey[8], with the Satterthwaite degrees of freedom of Pustejovsky and Tipton[10,14]. The CR2 adjustment and its associated small-sample test were designed for precisely the regime here, a moderate number of clusters, and are implemented for `lme` objects in the `clubSandwich` package[19,20].

It is worth setting down the construction. Let $\hat{e}_i$ collect the residuals of cluster i from the fitted model and let H_{ii} be the corresponding block of the hat matrix. A cluster-robust variance replaces the model-based variance of $\widehat{\theta}$ by a quadratic form in the empirical cluster cross-products,

$$\hat{V}_{\mathrm{CR}} \;=\; (X'X)^{-1}\Big(\sum_i X_i' A_i\, \hat{e}_i \hat{e}_i'\, A_i X_i\Big)(X'X)^{-1},$$

and the family CR0, CR1, CR2, CR3 is indexed by the adjustment matrices A_i[9,10]. The unadjusted choice $A_i = I$ is consistent as the number of clusters grows but biased downward at moderate cluster counts; the jackknife choice $A_i = (I - H_{ii})^{-1}$ over-corrects and is, in the marginal setting, the estimator of Mancl and DeRouen[7]. The CR2 choice sets $A_i = (I - H_{ii})^{-1/2}$, the symmetric square root that renders $\hat{V}_{\mathrm{CR}}$ exactly unbiased for the true variance when the working covariance is in fact correct; it is the cluster generalisation of the HC2 estimator of MacKinnon and White[21]. For a mixed model fitted by `lme` the construction is applied in the metric of the working covariance, that is to the generalized-least-squares-transformed design and residuals, following Pustejovsky and Tipton[10], and is recommended with a matching small-sample reference by Imbens and Kolesar[22]. Because $\hat{V}_{\mathrm{CR}}$ is itself a quadratic form in approximately normal residuals, its null distribution is approximated by a scaled χ^2, and the Satterthwaite degrees of freedom are recovered by matching its first two moments, $\nu = (\sum_g \lambda_g)^2 / \sum_g \lambda_g^2$ in the eigenvalues λ_g of the matrix defining that form[10,14].

Table 4 reports the verification. The model-based standard-error ratio reproduces the conservatism, between 1.05 and 1.10. The CR2 cluster-robust ratio is close to one, between 0.97 and 1.00, and the realised type-I error rises from approximately 0.03 under the model-based test to between 0.045 and 0.049 under the CR2 test, within Monte Carlo error of the nominal 0.05. The Satterthwaite degrees of freedom for the cluster-robust test are approximately 24, far below the model-based 487 and well below the participant count of 70, since they measure the heterogeneity of cluster leverage on the interaction contrast rather than a count of clusters[22]; the small-sample reference is doing real work, and a naive normal reference for the sandwich statistic would itself be mildly anti-conservative[9,23].

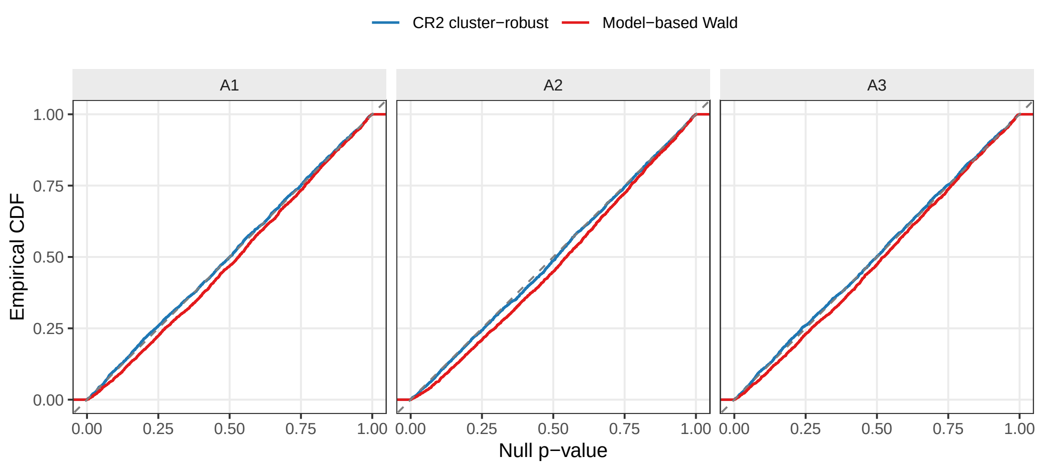

Figure 4: Null p-value distribution functions under the model-based and CR2 cluster-robust tests, by specification. The model-based curves bow below the diagonal; the CR2 curves lie on it.

Table 5: The cluster-robust remedy at an alternative cell (Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$, exponential DGP, 3000 replicates). Power is the rejection rate of the interaction test; the standard-error ratio r is as defined for the null. The model-based test is conservative here too, and the CR2 test recalibrates it.

Spec	r model-based	r CR2	Power model	Power CR2
A1	1.124	0.986	0.757	0.795
A2	1.152	0.988	0.843	0.878
A3	1.119	0.989	0.755	0.788

Figure 4 shows the same result as a distribution function. The model-based p-values bow below the diagonal, as in Figure 1; the cluster-robust p-values lie on it. The point estimate and the corCAR1 model are unchanged; only the standard error of the interaction, and the reference against which it is tested, are replaced.

4.5 The remedy at an alternative cell

The diagnosis to this point has been conducted under the null, where calibration is defined. A reader of a comparative study, however, will press a further question. The purpose of the parent study is to rank analysis specifications by power, and if a recalibration of the standard error raises the power of every specification, does it also disturb their order? We checked this directly at one alternative cell, the parent study's reference configuration with a non-null biomarker moderation, $c_{bm} = 0.45$, retaining the Architecture-B Hybrid design at $N = 70$.

Three things follow from Table 5. First, the standard-error over-estimation is not confined to the null: at the alternative cell the model-based ratio is 1.12 to 1.15, if anything slightly larger than the 1.06 to 1.10 seen under the null. Second, the CR2 cluster-robust standard error recalibrates the alternative cell as it did the null, bringing the ratio to 0.99 for all three specifications. Third, and most consequential for a comparative study, the recalibration is close to a uniform level shift. It raises the power of every specification by roughly three to four percentage points, and the gaps between specifications are left intact: the advantage of the leading specification A2 over A1 is +0.086 under the model-based test and +0.082 under the cluster-robust test, and over A3 it is +0.088 and +0.089. The ranking, and the magnitude of the leading specification's advantage, survive the recalibration unchanged.

The lesson generalises beyond the example. A cluster-robust standard error corrects the absolute error rate, the type-I error under the null and the power under the alternative alike, but because the correction acts on the standard error and not on the point estimate, and because the standard-error mis-calibration here is similar across the specifications being compared, it shifts the level without re-ordering the contrast. A comparative study whose claim is a ranking is therefore robust to the recalibration, even where its absolute power figures, as reported

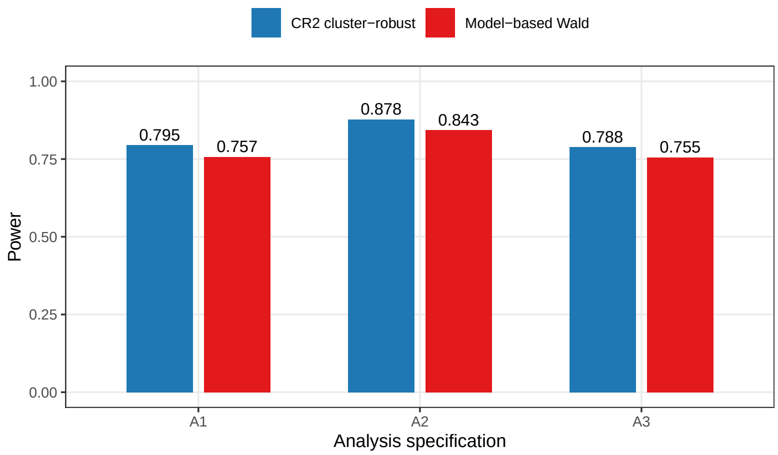

Figure 5: Power of the three analysis specifications at the alternative cell, under the model-based and CR2 cluster-robust tests. The recalibration raises every specification by a near-uniform increment and preserves their order.

under the model-based test, are mildly pessimistic.

5 Discussion

5.1 The result is not about autoregressive correlation

It would be a mistake to read this paper as a caution against the continuous-time autoregressive residual in particular. The corCAR1 term is the component the model ladder happened to implicate in this example, but the mechanism is general. The model ladder is itself the evidence: it fitted four working covariances and obtained four different standard-error ratios, from 1.3 under ordinary least squares to 1.0 under a random intercept to near 1.1 under the production model. The mis-calibration is a property of the match between the working covariance and the truth, and any parametric residual or random-effects structure, compound symmetry, an unstructured form, a random slope, a spatial correlation, is capable of producing it. That a mis-specified working covariance leaves the model-based fixed-effect standard error inconsistent is, moreover, not a new claim; it is the Eicker-Huber-White result carried into the mixed-model setting[3–5,24]. What the worked example contributes is a quantified, decomposed instance in a realistic longitudinal interaction test, with two features that are not obvious in advance.

5.2 Two features that are not obvious in advance

The first is the direction. The sandwich literature is usually motivated by anti-conservative tests: ignored positive correlation under-states standard errors, and the robust estimator corrects an inflated rejection rate downward. Here the model-based test is conservative, and the cluster-robust estimator corrects the rejection rate upward. The ladder explains why the intuition can mislead. The effect under test is a within-cluster contrast, and for a within-cluster contrast the relationship between a mis-specified covariance and the standard-error error need not have the sign that the between-cluster intuition supplies. An analyst who reasons that 'my standard errors are conservative because I have modelled the correlation' has the logic backward in this case: modelling the correlation, through the corCAR1 term, is what introduced the conservatism, and the less elaborate model was the calibrated one.

The second is that the bias is asymptotic. It would have been comfortable to attribute the conservatism to moderate sample size, and an earlier informal reading of the parent study did so. The sample-size gradient refutes that reading directly. This matters for how the problem is classified. The small-sample corrections of Kenward and Roger[15,16] adjust the fixed-effect standard error and its degrees of freedom for the finite-sample uncertainty in the estimated variance parameters, operationalising the variance-underestimation result of Kackar and Harville[25]; they are the right instrument for a finite-sample artefact. They are not the right instrument here, because the bias is not a finite-sample artefact: it does not vanish as the variance parameters are estimated with unlimited precision. A Kenward-Roger correction to a mis-specified working covariance corrects the finite-sample term and leaves the asymptotic mismatch untouched. The cluster-robust standard error addresses the asymptotic mismatch directly, which is why it is the appropriate remedy here.

5.3 When the problem should be expected

The conditions for the problem are not exotic. They are a fixed-effect interaction tested in a mixed model; a parametric working covariance that is plausible but not known to be exactly correct, which is to say almost every parametric working covariance in applied use; and an interaction that is partly or wholly a within-cluster contrast. Wherever those conditions hold, the calibration of the model-based Wald test is an open question, and the staged protocol of Section 2 answers it at the cost of a simulation. We would suggest, as a matter of routine practice, that an interaction test reported from a mixed model with a non-trivial working covariance be accompanied either by a calibration check of this kind or by a cluster-robust standard error, which removes the question by being agnostic to the working covariance. This accords with a broader case for giving robust and marginal methods more routine consideration alongside the conditional mixed model[26].

5.4 The marginal-model alternative

The remedy adopted above keeps the conditional mixed model and adds a cluster-robust standard error after the fact. A reader may reasonably ask whether the marginal route, generalized estimating equations, would not be more natural, since the method carries the sandwich variance from the outset rather than bolting it on[6]. It would. For a linear model with an identity link the marginal and conditional fixed-effect coefficients coincide, so a generalized estimating equations analysis estimates the same interaction and answers the same question; the working correlation, exchangeable or autoregressive, governs only the efficiency of the point estimate and not the validity of the inference.

We checked the marginal route at the primary null cell, fitting the generalized estimating equations counterpart of the analysis model with an AR(1) working correlation. The naive sandwich standard error proved mildly anti-conservative, the small-sample behaviour the literature describes for a moderate number of clusters: across the three specifications it under-estimated the sampling standard deviation by roughly four percent, and the realised type-I error was 0.067 to 0.069. A bias-corrected sandwich of the kind introduced by Mancl and DeRouen[7] removed most of the discrepancy, bringing the standard-error ratio to within one percent of one and the type-I error to 0.055 to 0.058. The marginal route is therefore viable, but it carries the same lesson as the conditional one: at the cluster counts typical of aggregated N-of-1 work the sandwich variance itself needs a small-sample correction. The two corrected tests err in opposite and modest directions, the conditional CR2 test slightly conservative and the marginal Mancl-DeRouen test slightly anti-conservative, and either is a defensible calibrated analysis. The choice between them is practical rather than principled: the conditional route preserves an existing mixed-model pipeline and changes only the standard error, while the marginal route is conceptually cleaner but a more thoroughgoing change of model.

5.5 Limitations

Several limitations bound the scope of the evidence. The diagnosis was carried out for one data-generating family, one estimand, and one production working covariance; the asymptotic conservatism is demonstrated for the corCAR1-plus-random-intercept model, and the generality beyond that rests on the model ladder and on established theory rather than on a sweep of every covariance structure. A broader empirical base, fitting an unstructured or compound-symmetry analysis to the same data, would strengthen the general claim, and is a natural extension. The cluster-robust remedy is not without cost: the Satterthwaite degrees of freedom for the CR2 test are approximately 24 against a model-based 487, so the robust test spends precision to buy validity, and at a genuinely small number of clusters that price would rise. The principal diagnosis was conducted at the null, where calibration is defined; the consequence for power was checked at a single alternative cell (Section 4.5), where the recalibration proved a near-uniform level shift that preserved the ranking of specifications. Whether that uniformity holds across the whole factorial, rather than at the one cell examined, was not established. Finally, the small positive point-estimate bias noted in Section 4.1, while negligible in magnitude and irrelevant to the conservatism, is a genuine finite-sample feature of the estimator that a separate examination might characterise.

6 Conclusions

1. The model-based Wald test of a fixed-effect interaction in a linear mixed model can be materially mis-calibrated when the parametric working covari-

ance does not match the data-generating covariance. In the worked example the mis-calibration is a conservative one: the model-based standard error of the interaction is over-estimated by 6 to 10 percent and the realised type-I error is approximately 0.03 against a nominal 0.05.
2. The four-channel decomposition localises the discrepancy to the standard-error channel and excludes point-estimate bias, the degrees-of-freedom rule, and the positive-definiteness correction.
3. The working-covariance model ladder localises the cause to the corCAR1 residual layer. A random intercept alone is well calibrated; adding the corCAR1 residual removes the calibration.
4. The sample-size gradient shows the mis-calibration is asymptotic. It does not attenuate from $N = 35$ to $N = 280$, and so it is a property of the working covariance and not a small-sample artefact amenable to a Kenward-Roger correction.
5. A CR2 bias-reduced cluster-robust standard error, clustered on participant and referred to a Satterthwaite distribution, restores the standard-error ratio to approximately one and the type-I error to approximately 0.05, without altering the point estimate or the analysis model. At an alternative cell the same recalibration is a near-uniform level shift in power that preserves the comparative ranking of specifications.
6. The staged protocol, the four-channel decomposition, the model ladder, and the sample-size gradient, is a reusable instrument for any mixed-model interaction test whose calibration is in doubt. We recommend that interaction tests reported from mixed models with a non-trivial working covariance be accompanied by a calibration check of this kind, or by a cluster-robust standard error that is agnostic to the working covariance.

7 Reproducibility

The diagnosis is driven by six scripts under `analysis/scripts/carryover-sensitivity/`: `diagnose-null-type1.R` for the four-channel decomposition, `diagnose-null-type1-stages46.R` for the model ladder and the sample-size gradient, `diagnose-null-type1-cr2.R` for the cluster-robust verification at the null, `diagnose-alt-cell-cr2.R` for the alternative-cell check, and `diagnose-null-type1-gee.R` for the GEE marginal-model check, with `check-null-architecture-equivalence.R` establishing the architecture-invariance of the null. Each writes a replicate-level `.rds` file into the `output/` subdirectory, and this manuscript reads those files and regenerates every table and figure inline. The cluster-robust standard errors use the `clubSandwich` package. The examination plan that preceded the manuscript is retained alongside it as `type1-examination-plan.md`.

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